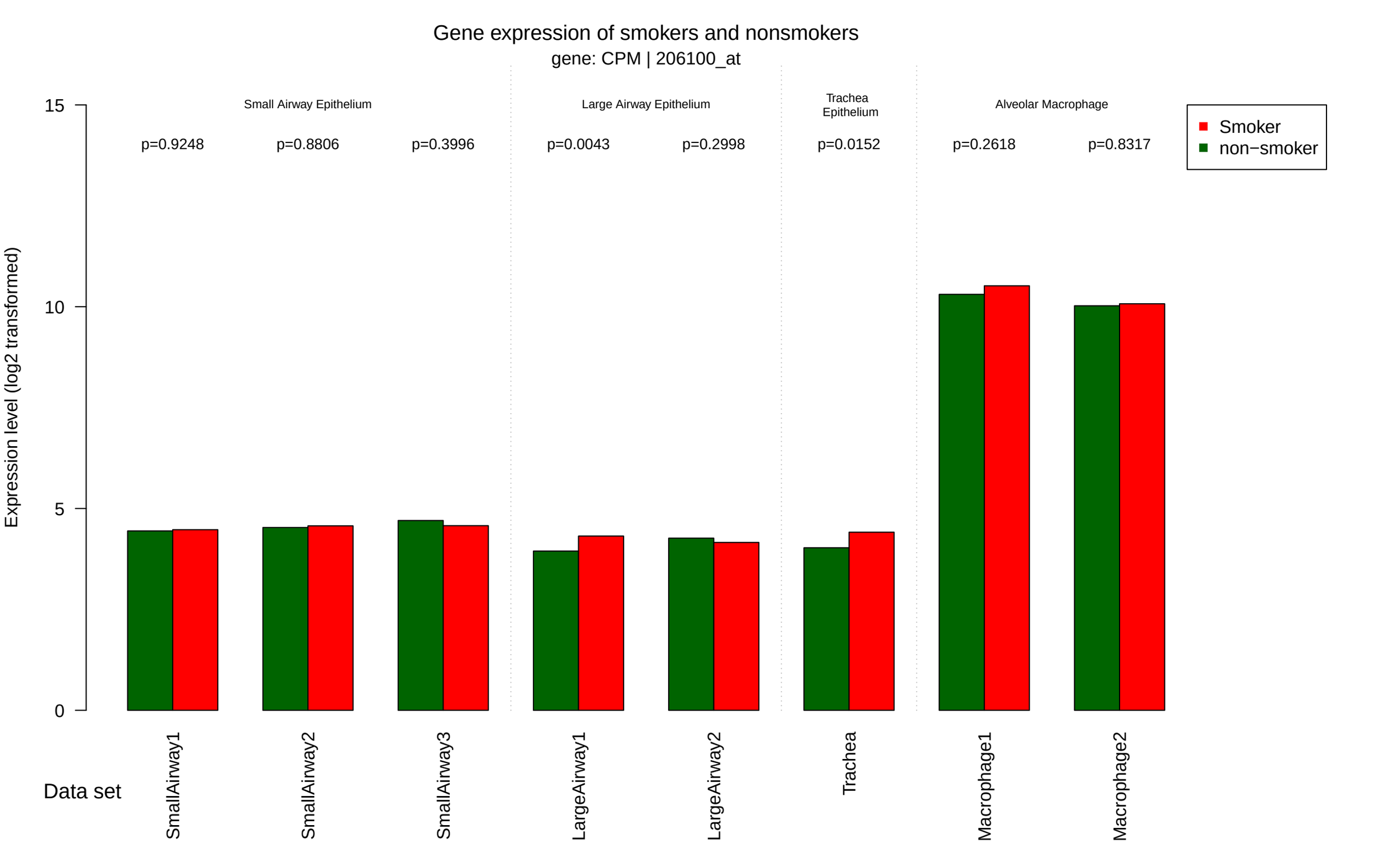
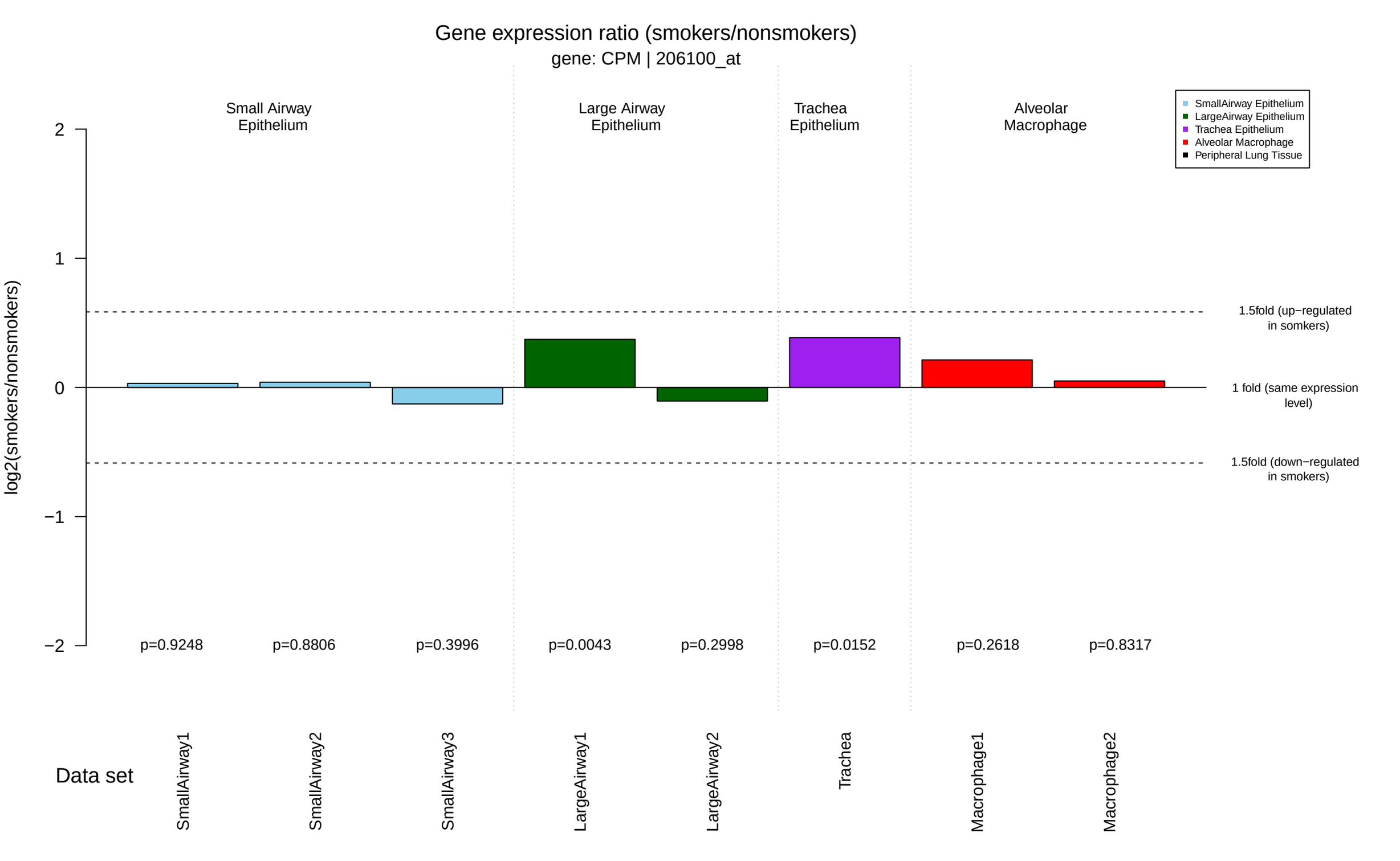
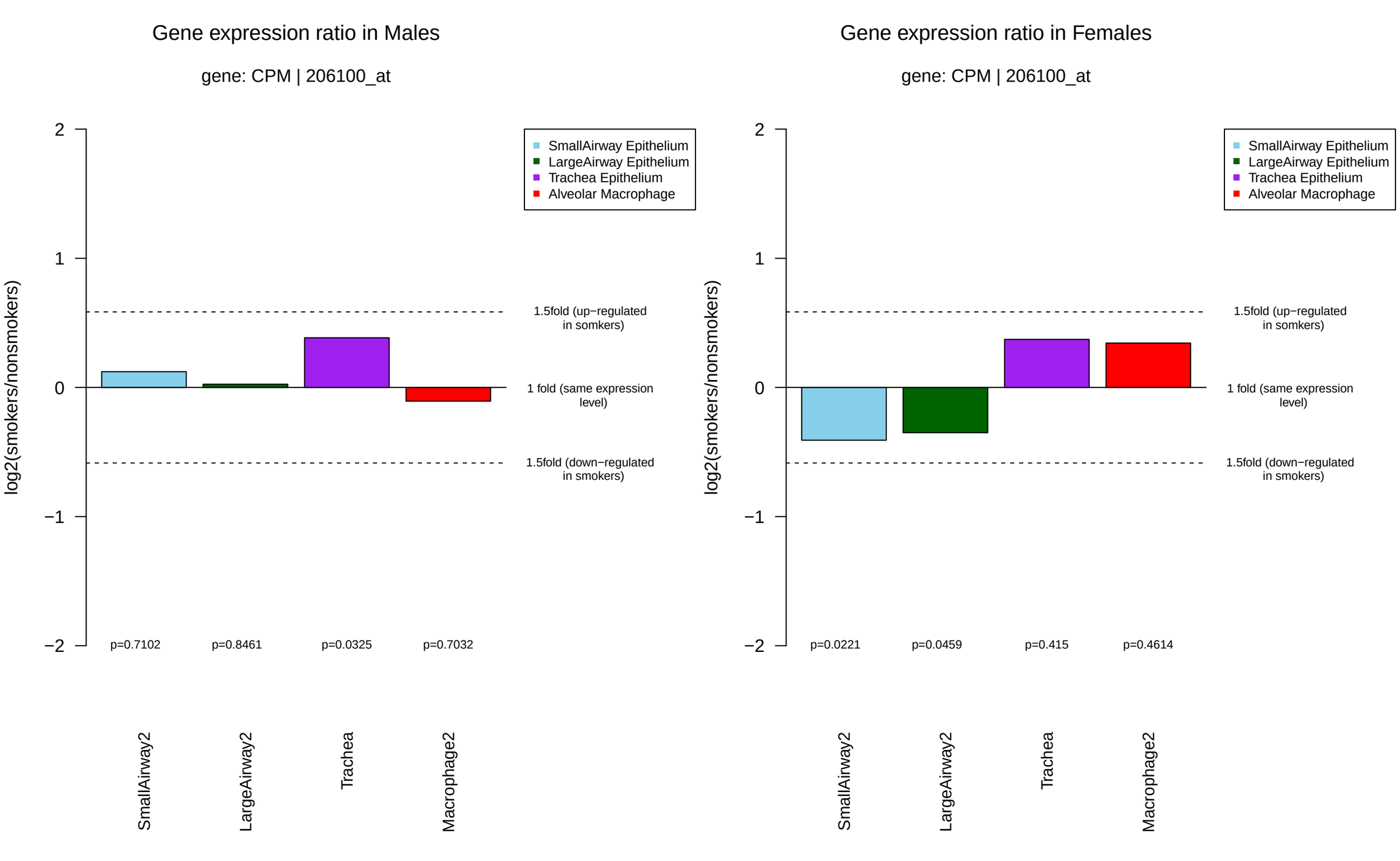
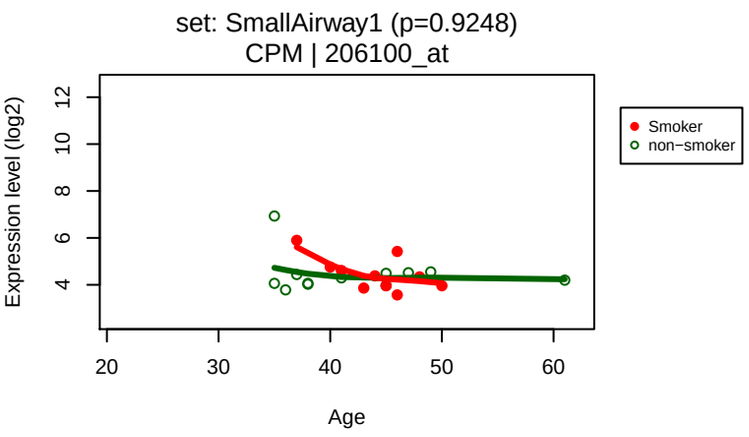
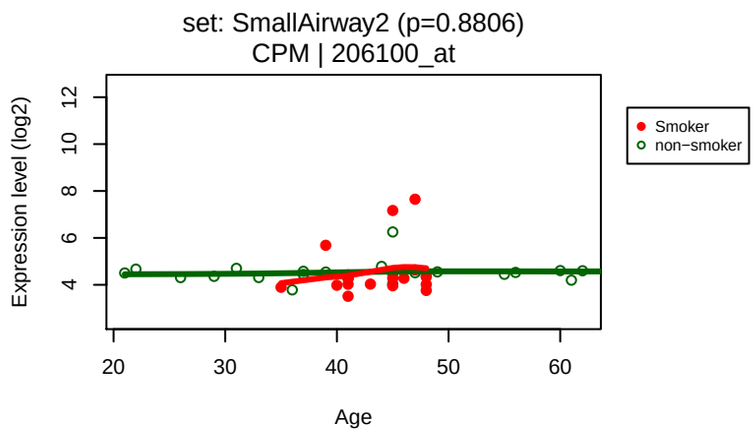
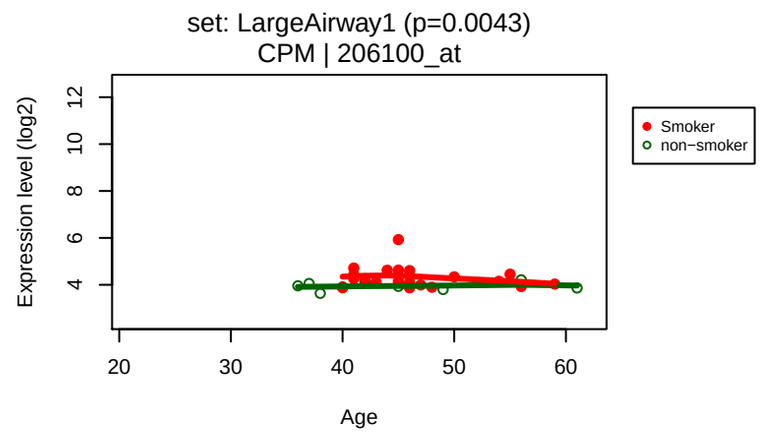

Gene expression of smokers and nonsmokers
gene: CPM | 206100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9248
p=0.8806
p=0.3996
p=0.0043
p=0.2998
p=0.0152
p=0.2618
p=0.8317
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

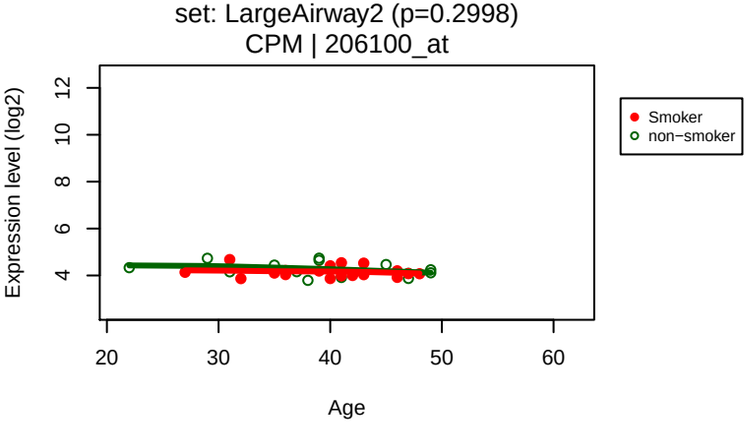

Gene expression ratio (smokers/nonsmokers)
gene: CPM | 206100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9248
p=0.8806
p=0.3996
p=0.0043
p=0.2998
p=0.0152
p=0.2618
p=0.8317
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

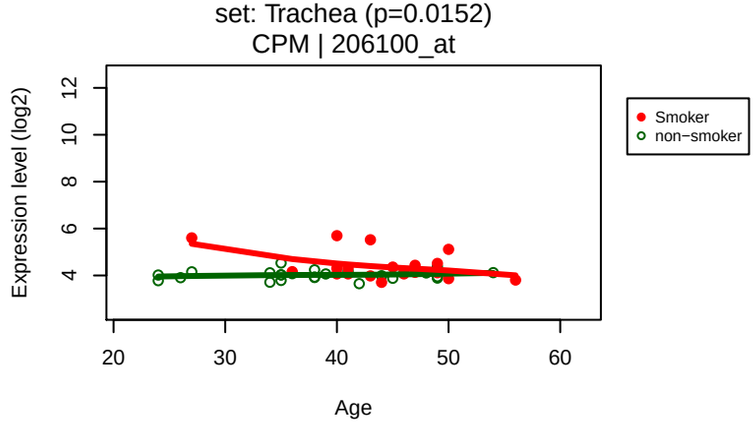

Gene expression ratio in Males
gene: CPM | 206100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7102
p=0.8461
p=0.0325
p=0.7032
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CPM | 206100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0221
p=0.0459
p=0.415
p=0.4614
SmallAirway2
LargeAirway2
Trachea
Macrophage2

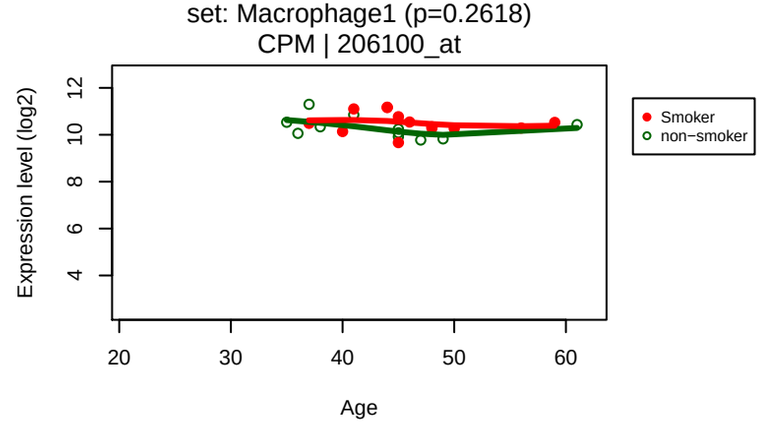

set: SmallAirway1 (p=0.9248)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker

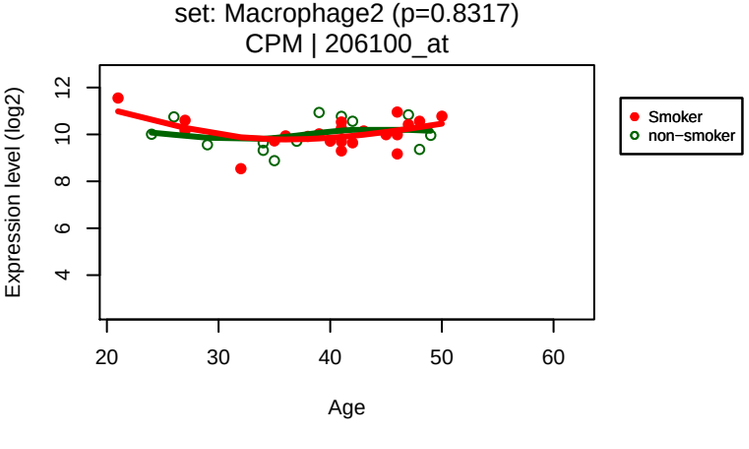

set: SmallAirway2 (p=0.8806)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker

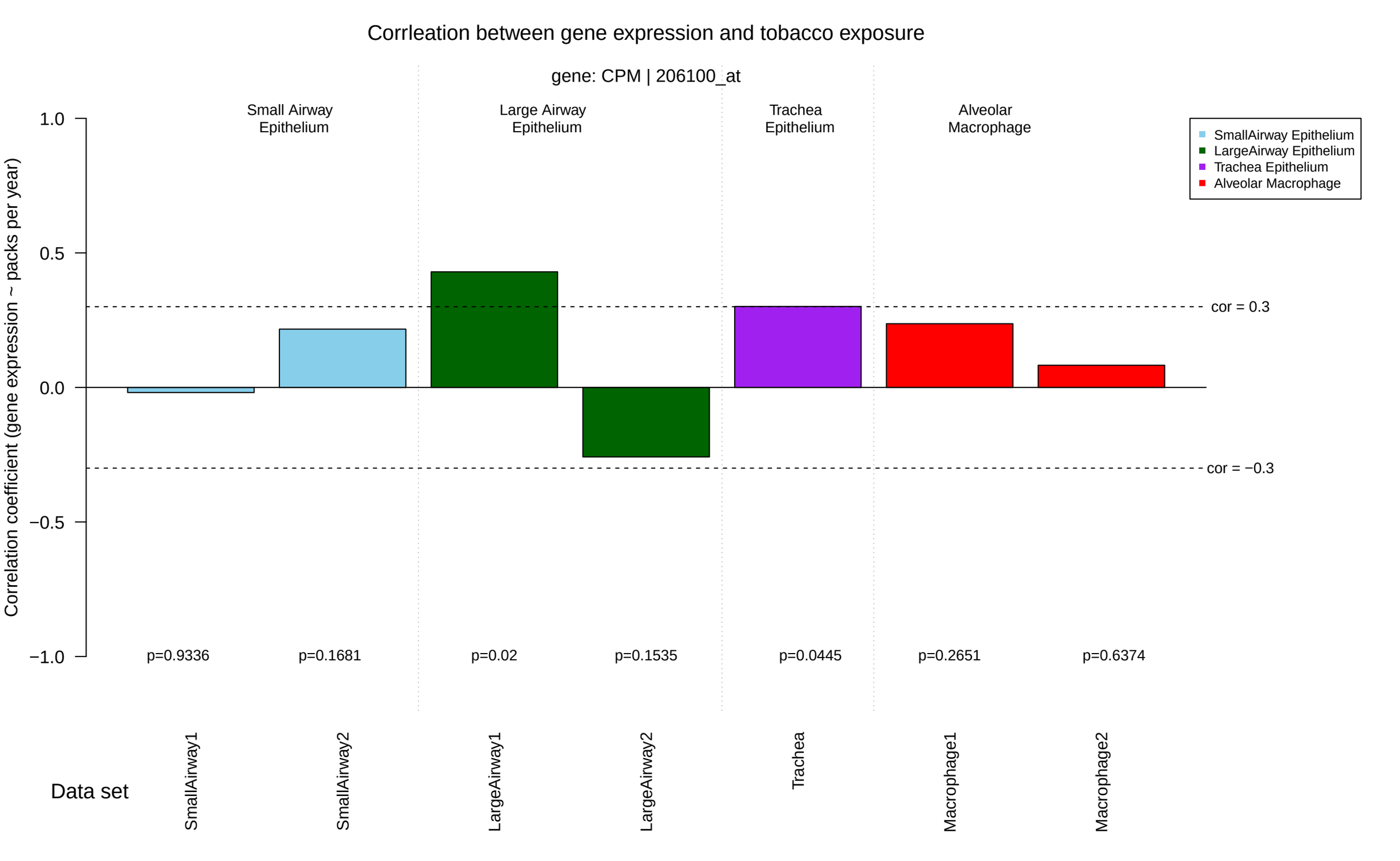

set: LargeAirway1 (p=0.0043)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2998)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0152)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2618)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8317)
CPM | 206100_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CPM | 206100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9336
p=0.1681
p=0.02
p=0.1535
p=0.0445
p=0.2651
p=0.6374
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2